MoodTrail-BD: A Longitudinal Mood-State-Labeled Social Media Resource for Bipolar Disorder

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Abstract. Bipolar disorder (BD) is frequently misdiagnosed as major depressive disorder (MDD), and monitoring mood state *transitions* over time is critical for timely intervention. Existing social media datasets for BD research typically provide only binary diagnostic labels or per-post mood scores, but none capture mood trajectories, i.e., the temporal dynamics of state transitions that define BD clinically. We present a longitudinal mood-state-labeled social media resource that provides annotations at two granularities: per-post categorical mood state classification and 14-day period-level mood trend analysis. The latter captures dominant mood state, trend direction (stable, worsening toward mania or depression, or fluctuating), and clinical change points within each period, enabling computational modeling of mood trajectories that per-post labels alone cannot support. The annotation schema is grounded in DSM-5 episode criteria and implemented as an LLM-based pipeline (Gemini 3.1 Pro). The dataset covers *TBD* self-identified BD users drawn from BD-focused subreddits (*TBD* periods, *TBD* submissions, *TBD* comments spanning *TBD*) with mood distributions consistent with known BD epidemiology. Post-level state classification is externally validated against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with high depressive recall (87.9%) and lower manic-pole recall (35.7%/6.7%). Six clinically significant error patterns are characterized, and we identify a structural label-text consistency issue at the manic pole that bounds achievable per-post manic recall.

Keywords: Bipolar Disorder · Large Language Models · Dataset Construction · Social Media · Clinical NLP · Mental Health

1 Introduction

Bipolar disorder (BD) is characterized by recurrent episodes of mania, hypomania, and depression, affecting 1–2% of the global population [2]. An estimated 17–50% of BD cases are initially misdiagnosed as major depressive disorder (MDD), because patients typically seek help during depressive episodes and may not recognize manic or hypomanic states as pathological [3, 4]. This misdiagnosis leads to inappropriate treatment (e.g., antidepressant monotherapy, which may trigger manic switching) and delays proper intervention.

Social media platforms such as Reddit host BD communities (e.g., r/bipolar, r/BipolarReddit) where users openly discuss symptoms, treatment, and daily functioning. Prior work has used these data for BD and MDD classification [5–7], but existing datasets provide only binary diagnosis labels (BD vs. MDD) or user-level risk scores. None offer post-level mood state labels tracked over time, which limits computational research on mood trajectories.

Recent work has applied LLMs to mental health NLP tasks [8, 9], but whether they can classify BD mood states at expert-level accuracy, especially manic-pole states that are difficult to detect from text, has not been established.

This paper presents MoodTrail-BD, a resource addressing this limitation. We collect Reddit posts from users who self-identify as having a BD diagnosis in BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2) and annotate them at two granularities: (1) per-post mood state (Depressive, Stable, Hypomanic, Manic) and (2) 14-day period-level mood trends (dominant state, trend direction, change points). The annotation schema follows DSM-5 episode criteria and is implemented as an LLM pipeline (Gemini 3.1 Pro). We validate the post-level annotations against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with 87.9% depressive recall and 35.7%/6.7% recall on hypomania/mania. The recall asymmetry across poles reflects both a known model property (manic-side states often manifest through described behaviors rather than affective tone) and a structural label-text consistency issue with the source dataset's manic-pole labels.

Our contributions:

1. A two-granularity annotation schema (per-post state + 14-day trend) with an LLM pipeline. Existing BD social media resources provide per-user or per-post labels; ours adds period-level mood trajectory annotations.
2. MoodTrail-BD comprises *TBD* self-identified BD users from BD-focused subreddits, *TBD* 14-day periods, *TBD* posts and comments spanning *TBD*, with mood and trend distributions consistent with clinical expectations.
3. External validation against the BD-Risk expert-labeled dataset on a held-out, author-disjoint, stratified evaluation subset, with macro F1 of 0.519 (87.9% depressive recall, 35.7% hypomanic recall, 6.7% manic recall) and six characterized failure modes.

2 Related Work

2.1 Longitudinal Mood Monitoring in BD

Clinical efforts to track BD mood trajectories rely heavily on ecological momentary assessment (EMA) and digital phenotyping with smartphone sensors and self-report apps [10, 11]. These approaches yield dense, high-frequency mood signals but require active patient enrollment and consent, limiting cohort size and external use. Social media offers a complementary unobtrusive source: it provides passive, naturally produced language data over months to years for users already discussing their condition. To date, however, mood trajectory annotations of social media data at the period level have not been released as a public resource.

2.2 Social Media in Mental Health Detection

De Choudhury et al. [12] showed that social media signals can predict depression onset. The SMHD dataset [5] covers nine mental health conditions via self-reported diagnoses; Coppersmith et al. [6] established shared tasks for depression and PTSD detection from Twitter.

For BD, Sekuli'c et al. [7] proposed Reddit-based classification and Jagfeld et al. [13] compiled a large BD Reddit corpus, but both rely on self-reported diagnoses without expert validation [14, 15]. The BD-Risk dataset [1] provides per-post mood labels validated by a psychiatrist and a clinical psychologist; we use it as the gold standard for our post-level validation.

2.3 LLMs for Clinical NLP and Mental Health

Xu et al. [8] evaluated LLMs on mental health prediction from online text; Yang et al. [9] fine-tuned MentalLLaMA for interpretable mental health analysis. Lee et al. [1] found that ChatGPT achieved only an F1 of 0.130 on BD risk detection (vs. 0.578 for their multi-task model), showing that off-the-shelf LLMs struggle with BD-specific tasks. On the more general question of LLM annotation quality, Gilardi et al. [16] show that ChatGPT can match or exceed crowd workers on text annotation tasks; our work follows this line by treating the LLM as an annotator (not a classifier) and grounding its outputs in an explicit DSM-5-derived schema.

Our work differs from these evaluations in that we do not predict diagnosis. Instead, we use an LLM to *annotate* per-post mood states and period-level trends, validate the annotations against expert labels, and release the result as a resource. The error analysis (Section 5) characterizes the failure modes that the schema must address and that downstream users should remain aware of.

3 Methodology

3.1 Resource Overview

MoodTrail-BD is a longitudinal mood-state-labeled Reddit corpus for BD research. It covers users who self-identify as having a BD diagnosis in BD-focused subreddits and provides annotations at two temporal granularities: (1) per-post mood state classification and (2) 14-day period-level mood trend analysis. Fig. 1 illustrates the construction pipeline.

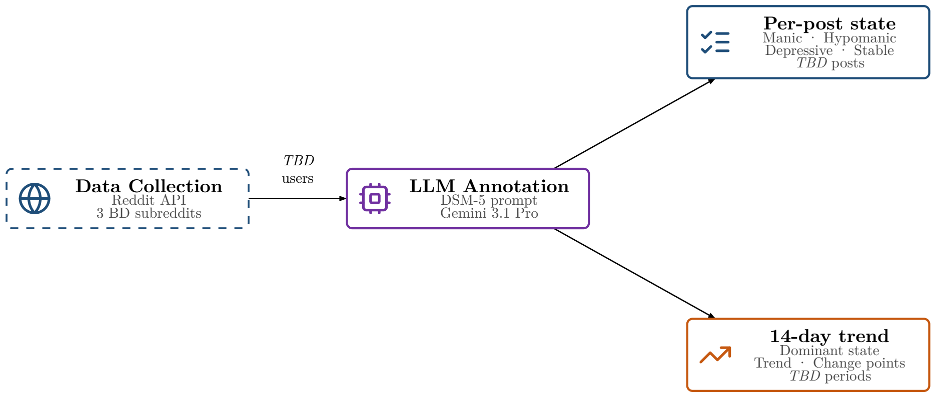

Fig. 1. MoodTrail-BD construction pipeline. Users are continuously collected from three BD-focused subreddits and filtered to those with sufficient longitudinal posting history. The selected cohort then undergoes a single LLM annotation step (Gemini 3.1 Pro, DSM-5 prompt) that yields labels at two temporal granularities.

Data collection. We continuously monitor three BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2) using the Reddit API. For each active author discovered, we retrieve their complete posting history (submissions and comments) and schedule periodic re-crawls with a configurable cooldown to capture ongoing activity.

Inclusion criteria. Users are included if they posted to at least one of the monitored BD-focused subreddits during the collection period, which serves as self-identification of BD relevance. We do not perform clinical verification of the diagnosis; this matches the inclusion model used in prior BD social-media datasets [7, 13]. The diagnostic status of individual users in the cohort is therefore self-reported rather than clinically confirmed (see Limitations).

Scale. MoodTrail-BD is built from *TBD* monitored users. Users with fewer than two 14-day periods containing posts (i.e., insufficient longitudinal span for trend analysis) are excluded. The remaining *TBD* users contribute *TBD* submissions and *TBD* comments spanning *TBD*, yielding *TBD* valid analysis periods.

3.2 Annotation Schema

Our annotation schema draws on DSM-5 episode definitions [17], operationalizing them into a structured prompting framework for LLM-based annotation. The schema described below was developed alongside an error analysis against external expert labels (see Section 5) that characterizes its known failure modes. It produces annotations at two temporal granularities:

Post-Level State Classification

For each individual post (submission or comment), the LLM assigns a categorical mood state from five options:

- **Manic:** Grandiosity, pressured writing (run-on sentences, excessive capitalization), flight of ideas (tangential topic shifts), extreme irritability or euphoria.
- **Hypomanic:** Elevated energy and pace with maintained coherence, social disinhibition, uncharacteristic intensity without psychotic features.
- **Depressive:** Linguistic constriction, absolutist language ("never," "nothing"), high self-focus (first-person pronouns), cognitive distortions, suicidal ideation.
- **Stable:** Balanced emotional tone, metacognitive reflection, proportionate responses, community-supportive language.
- **Uncertain:** Reserved for truly uninterpretable posts; the LLM must attempt classification before resorting to this label.

The framework also supports a `with_mixed_features` specifier (following DSM-5 mixed-features criteria). Before applying this specifier, the prompt requires the LLM to extract an explicit list of opposite-pole symptoms, and only assigns `with_mixed_features` when three or more clear opposite-pole symptoms are documented. This evidence-extraction step prevents mixed polarity from being used as a vague neutral label.

Period-Level Trend Analysis

For longitudinal mood trajectory modeling, we partition each user's posting history into consecutive fixed-length periods (default: 14 days). Periods are anchored at the user's first post (day 0) and advance in strict half-open intervals $[t_k, t_k + 14)$; submissions and comments are jointly assigned to the period containing their timestamp. All periods from the user's first to last post are defined; periods without posts receive a `NO_DATA` label rather than being skipped, preserving a continuous time grid for trajectory modeling. Fig. 2 illustrates the segmentation.

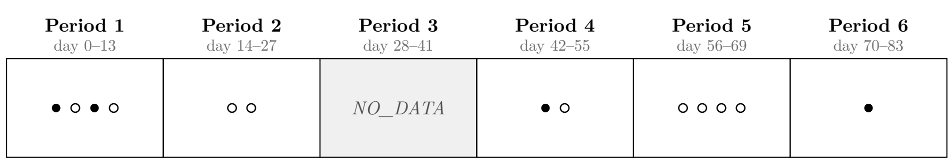

Fig. 2. Period segmentation for a single user (illustrative). Each box is a fixed 14-day window anchored to the user's first post (day 0). Submissions (filled circles) and comments (open circles) are jointly assigned to the period containing their timestamp. A period without posts (Period 3) is retained with a `NO_DATA` label rather than skipped, ensuring that downstream trajectory models see a continuous time grid.

For each period containing at least one post, the LLM analyzes the collected posts and produces:

- **Dominant state:** The primary mood state across the period (Manic, Hypomanic, Depressive, or Stable), using the same clinical criteria as post-level classification but aggregating evidence across multiple posts within the window.
- **Trend direction:** The trajectory of mood change during the period:
  - `NO_TREND`: the user maintains the same state throughout;
  - `TOWARDS_MANIA`: progressive worsening toward manic/hypomanic symptoms;
  - `TOWARDS_DEPRESSION`: progressive worsening toward depressive symptoms;
  - `FLUCTUATING`: alternation between states without a clear directional trend.
- **Change points:** Specific dates or events where a mood shift occurred, with pre- and post-shift states documented.
- **Trend summary:** A concise narrative describing the period's emotional trajectory and the evidence supporting the dominant state.
- **DSM-5 specifiers:** `with_mixed_features` when opposite-pole symptoms co-occur simultaneously within the period (as distinguished from sequential fluctuation).

Together, post-level states and period-level trends support event-level analysis (e.g., what preceded a state change) and longitudinal trajectory modeling (e.g., predicting the next period's dominant state from the current sequence). The explicit change-point fields additionally support change-point detection analyses [18] at the textual level.

3.3 External Validation Against BD-Risk

To establish the quality of the post-level state classification, we validate it against the clinically validated BD-Risk dataset introduced by Lee et al. [1] at NAACL 2024. Period-level trend analysis is not externally validated in this paper, as no existing dataset provides expert-annotated mood trajectories at the period level.

The BD-Risk Dataset

The BD-Risk dataset comprises 7,346 Reddit posts from 1,025 users collected from mental health subreddits (r/Depression, r/bipolar, r/BipolarReddit, r/BipolarSOs) spanning 2008 to 2023. Users are divided into two clinically defined groups: MDD-only (569 users) and MDD→BD (456 users, initially diagnosed with MDD and later re-diagnosed with BD). Because both groups are recruited via an initial MDD presentation rather than from active BD cohorts, the dataset is structurally enriched for depressive-pole content, with 88.9% of posts carrying mood labels ≤ 0. MoodTrail-BD, in contrast, samples from active BD-focused subreddits and preserves a broader cross-section of mood states across each user's longitudinal history.

Each post carries a mood level label on a 7-point scale (-3 to $+3$), assigned by trained annotators under the guidance of a psychiatrist and validated by two clinical experts (Krippendorff's $\alpha = 0.87$, inter-expert Cohen's $\kappa = 0.63$–0.69) [1].

Gold State Derivation

The BD-Risk dataset provides only ordinal mood labels; categorical states are not directly annotated. To obtain gold states for evaluation, we derive them from BD-Risk mood labels using the mapping shown in Table 1.

Table 1. Mapping from BD-Risk 7-point mood labels to derived gold states. This mapping treats +1 as within the normal positive range (Stable) rather than pathological activation.

BD-Risk mood label	Derived gold state	Notes
-3, -2, -1	Depressive	
0, +1	Stable	+1 = high motivation / positive mood within normal range
+2	Hypomanic	Clear manic-side activation without psychosis
+3	Manic	Severe manic expression with psychotic features

Evaluation Set Construction

The full BD-Risk dataset exhibits a heavily skewed mood distribution (88.9% of posts ≤ 0). Because the deployment task is BD risk detection rather than general mood classification, we deliberately oversample manic-pole posts so per-class metrics on the underrepresented classes carry meaningful statistical power.

We separate the labeled posts into two disjoint subsets. A *development* subset (314 posts) is used during prompt design and failure-mode analysis. A *held-out* subset (145 posts) is used exclusively for the evaluation reported below; it is **author-disjoint** from the development subset, drawn by stratified sampling from BD-Risk authors not present in the development subset, with quotas ensuring sufficient per-class support across the four derived gold states (60 Depressive, 40 Stable, 30 Hypomanic, 15 Manic). All metrics in Section 4.1 are computed on the held-out subset; the development subset is never used to produce reported numbers. Manic-pole gold posts in BD-Risk come almost exclusively from the MDD→BD group, so the held-out manic-side samples are structurally MDD→BD-derived (a limitation discussed in Section 6).

3.4 LLM Configuration

We use Gemini 3.1 Pro, accessed through the official API with structured JSON output. We selected this model for its large context window and native structured output generation, both required for the annotation pipeline. Each post is processed independently against a prompt containing the full annotation schema (DSM-5-grounded rules, output format) and eight synthetic few-shot examples targeting the failure modes characterized during prompt development (see Section 5). The prompt is supplied as a system instruction; the model returns a

JSON object with `state`, `opposite_pole_symptoms`, `specifiers`, `confidence` (High/Medium/Low), and `reasoning` fields. The `opposite_pole_symptoms` field carries the explicit evidence list required before the `with_mixed_features` specifier can be assigned (see Section 3.2). For period-level annotation, the LLM additionally returns `trend_direction`, `change_points`, and a `trend_summary` narrative, with a `confidence` value on a 0–1 scale rather than the post-level High/Medium/Low. Temperature is set to the default value. The few-shot examples are synthetic constructions written by the authors, not drawn from BD-Risk, and the model is not fine-tuned.

Gemini 3.1 Pro is the primary annotator throughout this paper. A cross-model probe with GPT-5.5 is reported in Section 4 to characterize schema portability and the impact of provider-level content-policy differences on annotation feasibility.

3.5 Evaluation Metrics

Throughout this paper, we refer to the expert-assigned labels from the BD-Risk dataset as gold labels and the LLM outputs as predictions. Gold states are derived from BD-Risk mood labels via the mapping in Table 1.

We report per-class precision, recall, and F1, along with overall accuracy and macro F1, plus the confusion matrix. Two accuracy variants are reported: accuracy *excluding* Uncertain treats Uncertain outputs as abstentions and removes them from both numerator and denominator (typical evaluator convention); accuracy *including* Uncertain counts Uncertain as incorrect, providing a conservative lower bound. Per-class precision, recall, and F1 follow the excluding-Uncertain convention. All metrics are computed on the full evaluation set.

4 Results

4.1 Post-Level Validation Against BD-Risk

We first validate the post-level state classification against BD-Risk expert labels. Table 2 presents per-class metrics; Table 3 reports overall agreement.

Table 2. Per-class state classification metrics on the held-out subset (n=145, excluding 7 Uncertain outputs). Manic precision is 1.0 because the single Manic prediction the LLM emitted was correct; recall (0.067) reflects that 14 of 15 gold-Manic posts were classified into another category.

State	Precision	Recall	F1	Support
Depressive	0.630	**0.879**	**0.734**	58
Stable	0.659	0.784	0.716	37
Hypomanic	**0.833**	0.357	0.500	28
Manic	**1.000**	0.067	0.125	15

Table 3. Overall state classification summary on the held-out subset (n=145). Macro F1 (rather than overall accuracy) is the primary metric because the held-out subset is intentionally stratified.

Metric	**Value**
Accuracy (excl. Uncertain)	65.9%
Accuracy (incl. Uncertain)	62.8%
Macro F1	0.519
Macro Precision	0.781
Macro Recall	0.522

Depressive recall is high (87.9%) and Stable recall is moderate (78.4%), while Hypomanic and Manic recall remain low (35.7% and 6.7%), indicating that the LLM correctly recognizes most depressive-pole and stable-pole posts but misses a majority of manic-pole cases. The confusion matrix (Table 4) makes the dominant error flow explicit: among the 30 gold-Hypomanic posts, 12 are predicted as Depressive and 6 as Stable; among the 15 gold-Manic posts, 11 are predicted as Depressive and 2 as Stable. The Manic-to-Depressive error flow is a recurring pattern with a likely label-text origin that we discuss in Section 6.

Table 4. State confusion matrix on the held-out subset (rows: derived gold state; columns: LLM prediction). DEP = Depressive, STA = Stable, HYP = Hypomanic, MAN = Manic, UNC = Uncertain.

Gold \ Pred.	**DEP**	**STA**	**HYP**	**MAN**	**UNC**	**Total**
Depressive	**51**	7	0	0	2	60
Stable	7	**29**	1	0	3	40
Hypomanic	12	6	**10**	0	2	30
Manic	11	2	1	**1**	0	15

These Hypomanic errors are concentrated on gold +2 posts whose manic-side activation is masked by negative tone, a pattern the prompt still struggles with.

4.2 The Uncertain Label as Quality Control

The LLM outputted Uncertain for 7 posts (4.8%), abstaining when post content was insufficient for state assessment. The non-zero rate reflects the prompt's explicit instruction to abstain on truly uninterpretable content rather than force a best-effort guess.

The Uncertain label serves as a quality filter preventing forced labeling when evidence is insufficient. The Uncertain emissions in the held-out evaluation are distributed across gold classes (2 Depressive, 3 Stable, 2 Hypomanic, 0 Manic), with no strong concentration on a single pole.

4.3 Schema Contribution: Comparison with a Zero-Shot Baseline

To quantify how much the structured annotation schema contributes beyond the LLM's base capability, we re-evaluate the same Gemini 3.1 Pro model on the held-out subset with a minimal zero-shot prompt: task definition only (post $\rightarrow$ one of five states), no DSM-5 rules, no SAFETY OVERRIDE, no Severity Descriptors, and no few-shot examples. The output JSON fields are unchanged so the comparison is apples-to-apples. Table 5 reports the side-by-side metrics.

Table 5. Schema contribution on the held-out subset (n=145). Both runs use the same model (Gemini 3.1 Pro) and the same output JSON fields; the only variable is the system prompt. The zero-shot prompt is a minimal task definition with no clinical rules; the full schema includes SAFETY OVERRIDE, Severity Descriptors, Clinical Guidance sections, and eight synthetic few-shot examples.

Metric	Zero-shot	Full schema	Δ
Accuracy (incl. Uncertain)	51.7%	**62.8%**	+11.1 pp
Accuracy (excl. Uncertain)	63.6%	65.9%	+2.3 pp
Macro F1	0.459	**0.519**	+0.060
Uncertain count	27	**7**	−20
DEPRESSIVE F1	**0.738**	0.734	−0.004
STABLE F1	0.600	**0.716**	+0.116
HYPOMANIC F1	0.500	0.500	± 0.000
MANIC F1	0.000	**0.125**	+0.125

Three observations frame the schema's value. First, the largest single effect is a 4× reduction in Uncertain emissions (27 $\rightarrow$ 7): the structured schema gives the LLM the vocabulary to commit to a label rather than abstain, which explains why the accuracy gain **including** Uncertain (+11.1 pp) is much larger than the gain **excluding** Uncertain (+2.3 pp). Second, STABLE classification benefits the most in per-class F1 (+0.116), driven by the Severity Descriptors' explicit "STABLE includes mild positive activation" rule that prevents the LLM from defaulting non-pathological positive posts to DEPRESSIVE or UNCERTAIN. Third, DEPRESSIVE and HYPOMANIC F1 are essentially unchanged, suggesting the LLM already handles those categories competently from base capability alone, and MANIC remains poorly recalled in both runs (0/15 zero-shot vs. 1/15 schema, counting Uncertain as incorrect), confirming that the manic-pole limitation is structural (label-text consistency, Section 6) rather than something a richer prompt can repair.

The comparison establishes that the schema's primary contribution is **coverage and decisiveness** (preventing abstention, anchoring border-class decisions) rather than raw classification accuracy on the cases the LLM is already confident about.

4.4 Cross-Model Annotation Feasibility

To probe whether the schema generalizes beyond Gemini, we additionally evaluated the full schema with OpenAI's GPT-5.5 (`reasoning_effort = "high"`) on the same 145 held-out posts. The schema and JSON output format were identical to the main run; only the underlying model changed.

Refusal under content policy. GPT-5.5 declined to classify 103 of 145 held-out posts (71.0%) with a content-policy refusal — the verbatim response was `"I'm sorry, but I cannot assist with that request."` — and produced no structured output for those posts. The refusals concentrated on posts containing explicit self-harm or suicidal content, exactly the safety-relevant subset that BD-Risk includes by clinical design and that the SAFETY OVERRIDE rule is built to handle. The prompt's clinical-research framing (the schema opens with "You are an expert AI assistant specializing in linguistic analysis for psychiatric research") did not lift the refusal. Gemini 3.1 Pro produced a structured output for all 145 posts under the same prompt.

Performance on the non-refused subset. On the 42 posts GPT-5.5 did classify, both models score higher than on the full 145, because the subset is skewed away from depressive-crisis content (gold distribution within the subset: 16 Depressive, 10 Stable, 12 Hypomanic, 4 Manic). Table 6 reports the head-to-head metrics on this subset.

Table 6. Cross-model comparison on the 42 posts GPT-5.5 accepted (out of 145 held-out). GPT-5.5 reaches a higher macro F1, driven mostly by Manic F1 (2 of 4 correct vs. 1 of 4 for Gemini). The result is **not** a clean head-to-head: the 42-post subset is the post set that survived GPT-5.5′s content-policy filter, which selectively excluded depressive-crisis posts. The remaining 103 posts (71% of the held-out set) are unlabeled by GPT-5.5 and excluded from this table.

Metric (on 42 non-refused posts)	**Gemini**	**GPT-5.5**
Accuracy (excl. Uncertain)	72.5%	**73.2%**
Macro F1	0.649	**0.710**
Macro Precision	0.827	**0.838**
Macro Recall	0.656	**0.679**
DEPRESSIVE F1	**0.800**	0.769
STABLE F1	0.727	**0.737**
HYPOMANIC F1	0.667	0.667
MANIC F1	0.400	**0.667**
Uncertain emissions	2	**1**

Interpretation. Three things follow. First, when GPT-5.5 does respond, the schema produces sensible structured output, confirming that it is not Gemini-specific in spirit. Second, the head-to-head numbers favor GPT-5.5 on the non-refused subset, but the subset selection itself is the dominant effect: GPT-5.5 systematically filtered out the hard depressive-crisis posts that drive most of

Gemini's error rate on the full held-out, so a direct macro-F1 comparison overstates GPT-5.5′s effective competence on a clinical mental-health corpus. Third, and most important in practice, production safety filters can render an LLM unsuitable as an annotator for psychiatric corpora: a 71% refusal rate on the held-out subset makes GPT-5.5 infeasible as a stand-alone annotator for MoodTrail-BD under its current safety policy, regardless of intrinsic capability. Choice of LLM provider has annotation-feasibility consequences beyond accuracy.

4.5 Author-Level Aggregation of Per-Post Predictions

A common downstream use of per-post labels is to aggregate them temporally, for example by taking the dominant state across the posts a user wrote in a 14-day window. To probe how much per-post errors smooth out under such aggregation, we compute author-dominant predictions on the held-out subset: for each author, the predicted state is the majority over their post-level LLM predictions (excluding Uncertain), and the gold state is the majority over their BD-Risk post-level gold labels. The 145 held-out posts come from 120 unique authors. Table 7 summarizes the resulting metrics.

Table 7. Per-post versus author-level (majority-vote) metrics on the held-out subset. Author-dominant aggregation produces a modest accuracy gain (most visible on the 20 evaluable authors with ≥ 2 posts, where it recovers $\simeq$5 percentage points over per-post). Macro F1 does not improve, and Manic recall remains essentially unchanged — aggregation smooths random per-post noise but does not repair the class-specific manic-side failure mode discussed in Section 6.

Metric	Per-post (n=145)	Author-dominant (n=120)
Accuracy (excl. Uncertain)	65.9%	**66.7%**
Macro F1	0.519	0.485
Manic recall	6.7% (1/15)	7.1% (1/14)
Authors with ≥ 2 posts: per-post acc	54.8% (23/42)	—
Authors with ≥ 2 posts: author-dominant acc	—	**60.0% (12/20)**

This pattern — modest aggregation gain on accuracy, no improvement on Manic recall — suggests that downstream temporal-aggregation systems can safely consume the per-post outputs and recover some of the per-post noise, but the residual manic-pole underdetection is not a noise property and will persist under any aggregation that operates on the same per-post inputs.

4.6 Resource Annotation: Period-Level Mood Trends

With post-level reliability confirmed, we turn to the period-level trend annotations. Fig. 3 shows the two-granularity annotation for two representative users: each row is a user, colored bars represent 14-day periods (color = dominant state, hatch pattern = trend direction), and dots represent individual post-level state annotations.

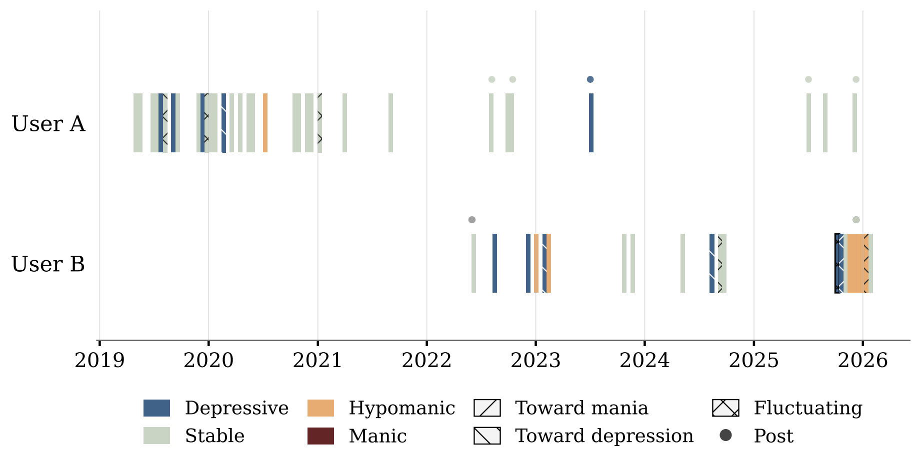

Fig. 3. Mood trajectory visualization for two anonymized BD users with multi-year posting histories. Colored bars represent 14-day periods (bar color = dominant state; hatch pattern = trend direction; the `with_mixed_features` specifier is drawn with a thin black border). Dots above each bar represent individual post-level state annotations, color-coded by post state.

Table 8 presents the distribution of trend directions across the annotated periods.

Table 8. Distribution of 14-day period trend directions (n = *TBD* periods from *TBD* users). The predominance of NO_TREND is clinically expected: most 14-day windows capture a single ongoing episode or stable phase rather than a state transition.

Trend direction	**Periods**	**%**	**Interpretation**
NO_TREND	*TBD*	*TBD*%	Mood state maintained throughout period
FLUCTUATING	*TBD*	*TBD*%	Alternation between states, no clear direction
TOWARDS_DEPRESSION	*TBD*	*TBD*%	Progressive worsening toward depressive pole
TOWARDS_MANIA	*TBD*	*TBD*%	Progressive worsening toward manic pole

Table 9. Distribution of dominant states across 14-day periods. The `with_mixed_features` specifier was applied to *TBD* periods (*TBD*%).

Dominant state	Periods	%
Stable	*TBD*	*TBD*%
Depressive	*TBD*	*TBD*%
Hypomanic	*TBD*	*TBD*%
Manic	*TBD*	*TBD*%

Most 14-day windows show NO_TREND, which is expected: a typical BD mood episode lasts weeks to months (depressive) or 1–4 weeks (manic) per DSM-5 [17], so state transitions within a single 14-day window are infrequent. The TOWARDS_DEPRESSION and TOWARDS_MANIA trends, though rare, mark episode onset or escalation, which are the signals most relevant for early intervention.

Stable and Depressive states dominate the dataset; manic-pole states (Hypomanic and Manic combined) account for the remainder. This matches the known BD asymmetry: depressive episodes are more frequent and longer, and users may post more during depressive and stable periods [13].

5 Error Analysis

This section characterizes the six failure modes that the schema must contend with, identified through manual analysis of disagreements between the LLM's predictions and the BD-Risk gold labels. Each pattern is named by its surface cause, illustrated with a representative case, and tied to the schema rule that constrains it. Manic-or-hypomanic posts misclassified as depressive or stable remain the dominant residual failure mode under the schema; the held-out per-class numbers in Section 4.1 quantify how often it occurs.

5.1 Pattern 1: Retrospective Behavioral Cues Ignored

The dominant manic-side error pattern. When users describe manic-episode behaviors (impulsive spending, aggressive confrontations, hyperactivity) in a retrospective post written with remorse or self-blame, the LLM assigns depressive labels based on the *current emotional tone* rather than recognizing the *clinical significance of the described behaviors.*

Representative case (synthetic): A user recounts a week-long episode of reckless spending and impulsive decisions, writing in a tone of deep regret and self-condemnation. The gold state is Hypomanic, recognizing the described behaviors (impulsive spending, reduced inhibition) as hallmark manic symptoms. The LLM predicts Depressive, anchoring on the self-deprecating narrative tone. The root cause: the LLM processes text as sentiment rather than as clinical evidence, conflating the *affect of the narrator* with the *clinical state being described*.

5.2 Pattern 2: Mixed Features Conflated with Neutrality

When posts exhibit symptoms from both poles simultaneously (e.g., severe sleep disruption and inability to concentrate alongside aggressive outbursts), the LLM sometimes defaults to Stable, treating the coexistence of opposing signals as cancellation rather than as a clinically significant mixed episode. Under DSM-5 criteria [17], mixed features should be annotated as a specifier on the primary state, with the state reflecting the dominant pole.

Representative case (synthetic): A user writes that they have not slept in three days, cannot focus at work, and snapped at their partner over a trivial misunderstanding, all in the same post. The gold state is Hypomanic with mixed features (decreased need for sleep alongside dysphoric irritability). The LLM predicts Stable, reasoning that the negative and positive signals "balance out." The root cause: the LLM treats opposite-pole symptoms as cancelling rather than co-existing under a mixed-features specifier.

5.3 Pattern 3: Calm Writing Style Masking Suicidal Ideation

A safety-critical pattern. Some posts express suicidal ideation ("I want to die") in calm, reflective, or educational prose. The LLM reads the linguistic register as stable and classifies accordingly. The gold state is Depressive (derived from expert mood label −3). This pattern highlights that calm writing does not rule out suicidal crisis, and any deployed system must detect crisis-level language regardless of surrounding tone. The schema addresses this with an explicit safety-override rule.

Representative case (synthetic): A user posts a coherent, structured essay reflecting on common public misconceptions about depression. Embedded inside the second paragraph, in the same calm register, is a single sentence noting that the writer themselves quietly thinks about not waking up most mornings. The gold state is Depressive (−3). Without the safety-override rule, the LLM weights the essay's overall register and classifies the post as Stable.

5.4 Pattern 4: End-of-Post Hope Overriding Pervasive Impairment

Posts describing severe functional impairment (academic collapse, inability to maintain daily routines, social withdrawal) sometimes conclude with a single hopeful statement (e.g., "my therapist said maybe I shouldn't give up"). The LLM anchors on this terminal positive signal and classifies the post as Stable, while the gold label reflects the pervasive impairment evidence throughout the post. This suggests a recency bias in the LLM's text processing.

Representative case (synthetic): A user reports having missed every class for the past month, eaten almost nothing this week, and lost the ability to reply to friends' messages, and closes the post with one line saying that maybe tomorrow will be different. The gold state is Depressive based on the pervasive functional impairment described throughout. The LLM anchors on the terminal optimism and predicts Stable.

5.5 Pattern 5: Substance-Induced Activation vs. Endogenous Mood

When users describe mood elevation caused by substances (e.g., caffeine-induced euphoria described as feeling “on top of the world”), the LLM may interpret these descriptions as evidence of hypomanic mood. The gold labels distinguish between pharmacologically induced activation and endogenous mood states, reflecting the user’s baseline mood rather than transient substance effects. The schema addresses this boundary with an explicit substance-mood disambiguation rule that also covers the recurrent-pattern case in which an unusual reactivity to a common substance is itself bipolar-spectrum.

Representative case (synthetic): A user writes that after their fourth coffee of the day they suddenly feel unbeatable and ready to overhaul their entire apartment, attributing the surge explicitly to caffeine and noting that they expect to crash by evening. The gold state is Stable (or Depressive baseline) because the activation is acute, externally caused, and consciously labeled as substance-driven. Without the rule, the LLM picks up the surface “unbeatable / ready to overhaul” cues and predicts Hypomanic.

5.6 Pattern 6: Uncertain Masking Embedded Clinical Signals

In rare cases, the LLM correctly identifies a post as metacommentary or informational but fails to recognize clinically significant content embedded within it. A post critiquing the “aestheticization of depression” online may contain explicit suicidal statements that are masked by the Uncertain classification. Uncertain functions appropriately in the large majority of cases, but the schema mitigates this specific pattern with the safety-override rule, which forces Depressive classification when crisis-level language is present, regardless of overall post framing.

Representative case (synthetic): A user writes a meta-commentary post criticizing how recovery is portrayed on social media, citing examples and arguing the framing is harmful. In the middle of the argument the writer notes, parenthetically, that they have been thinking about ending things themselves. Without the safety-override rule, the LLM classifies the post as Uncertain (because the dominant register is meta-discussion, not first-person mood), masking the embedded crisis-level signal. The gold state is Depressive.

6 Discussion

6.1 Period-Level Trends

Prior BD datasets provide per-user labels [5, 7] or per-post scores [1], but not mood *trajectories*. Our period-level annotations add this dimension: each 14-day window records not only the dominant state but also whether the user's mood is shifting (trend direction) and when shifts occur (change points).

From the trend distribution (Section 4.6), four observations stand out. (1) TOWARDS_MANIA and TOWARDS_DEPRESSION trends are rare but clinically the most important signals, as they mark episode onset where intervention has the most impact. (2) FLUCTUATING periods may correspond to rapid cycling or mixed presentations that single-post labels cannot capture. (3) Post-level states and period-level trends together enable hierarchical modeling: predicting the next period's trajectory from the sequence of post-level features. (4) The dominant-state distribution (Table 9) preserves meaningful manic-pole representation, in contrast to MDD-recruited cohorts such as BD-Risk where 88.9% of posts fall in the depressive-or-neutral range. This balance is important for downstream modeling work that must learn to distinguish manic-pole from depressive states rather than predicting the depressive majority class.

No existing dataset provides expert-annotated period-level mood trajectories, so we cannot externally validate the trend labels. The distributions are consistent with clinical expectations (see Section 4.6), but expert validation of trend annotations is needed in future work.

A complementary direction is to model the signal currently missing from our schema: periods with `NO_DATA` (see Fig. 2) are excluded from mood-state annotation because no text is available. Reduced social media activity is sometimes associated with depression in digital-phenotyping research [10], but the relationship is heterogeneous — some users post more during depressive episodes (rumination, support-seeking), others post less, and many non-posting periods reflect factors unrelated to mood (changing platform interest, offline life events, account suspension). Treating absence as a mood signal would therefore require posting-frequency features benchmarked against per-user baseline activity, and is beyond the scope of a text-based annotation pipeline.

6.2 Manic-Pole Underdetection

The LLM achieves 87.9% recall for Depressive states but only 35.7% for Hypomanic and 6.7% for Manic on the held-out subset. Two factors compound to produce this pattern. The first is a model property: depressive language has stereotypical surface markers (negativity, self-focus, hopelessness), while manic-pole states often manifest through *described behaviors* (spending sprees, reduced sleep need, grandiose plans) narrated in any tone (regret, confusion, or humor). The LLM reads tone rather than recognizing the clinical significance of the described behaviors (see Pattern 1 in Section 5).

The second factor is a label-text consistency issue rooted in the BD-Risk annotation rule. As specified by [1] (Section 3.2), "posts exhibiting both manic and depressive moods are regarded as manic moods," an asymmetric tie-breaking rule that elevates any post with mixed manic+depressive content to the manic side of the scale. In combination with the dataset's selection of MDD→BD users, this rule yields a non-trivial number of gold-label Manic (+3) posts whose textual content matches BD-Risk's own definition of -3 ("extreme anxiety and having suicidal thoughts"). A single-post LLM applying clinical-safety priors — classifying explicit self-harm content as Depressive — will systematically underperform on these specific posts, since the only signal the model has is the very content that the labeling rule overrode. The Manic-recall ceiling of 6.7% in Table 2 reflects this structural mismatch, not solely a model limitation. We discuss the implications for downstream use in Limitations.

7 Limitations

MoodTrail-BD is subject to several constraints, which we group into evaluation, annotation, and resource-construction limitations.

Limited cross-model coverage. The main BD-Risk validation reports a single annotator (Gemini 3.1 Pro). A complementary cross-model probe with GPT-5.5 (see Table 6) shows that the schema is not Gemini-specific in spirit but cannot be applied at scale under OpenAI's current content policy (71% refusal rate on the held-out subset). Other model families (Anthropic Claude, open-weight reasoning models) remain unexplored; the reported per-class behavior on the full 145-post held-out should still be read as Gemini-specific.

Post-level rather than longitudinal evaluation. Our quantitative validation is at the post level, whereas clinical mood assessment typically considers longitudinal patterns. The post-level setting was chosen because BD-Risk is the only existing dataset with per-post expert mood labels at sufficient scale; period-level trends, which are the resource's main contribution, are not externally validated because no external longitudinal ground truth exists at the period level.

Gold-state derivation, not direct expert annotation. Most importantly for interpreting the evaluation results, gold states are derived from BD-Risk mood labels via a deterministic mapping (Table 1) rather than being directly annotated by experts as categorical states. This mapping introduces several sources of imprecision: (a) the boundary between adjacent categories is inherently uncertain (a post with BD-Risk mood label $+1$ may reflect genuine mild hypomania rather than stable mood, yet the mapping deterministically assigns it to Stable); (b) the BD-Risk mood label itself is an ordinal intensity score, not a categorical clinical judgment, and the two dimensions do not always correspond one-to-one (for example, a recovery-narrative post within an ongoing depressive episode might receive a neutral or mildly positive mood score from an expert while remaining clinically depressive); (c) inter-expert agreement on the BD-Risk 7-point scale (Krippendorff's $\alpha = 0.87$) is measured at the ordinal level, but mapping this to categorical states amplifies disagreement at category boundaries. As a result, some apparent misclassifications in our evaluation may reflect mapping artifacts rather than genuine LLM failures, and the reported accuracy figures should be interpreted as lower bounds on the schema's true reliability.

Underrepresented manic pole. The held-out subset (145 posts) was constructed to give each derived gold state enough support for per-class metrics, but Manic ($n = 15$) remains small in absolute terms because the entire BD-Risk dataset contains only 28 mood-+3 posts. With $n = 15$, the 95% confidence interval around the Manic-F1 estimate is wide ($\pm$ approximately 13 percentage points), and small Manic-F1 differences should not be interpreted as significant. Increasing Manic statistical power would require re-annotation of additional sources, which is outside the scope of this work.

Label-text consistency on the Manic pole. The BD-Risk dataset specifies that "posts exhibiting both manic and depressive moods are regarded as manic moods" [1], an asymmetric tie-breaking rule that can yield gold-label Manic posts whose surface text matches the dataset's own definition of -3 (extreme anxiety with suicidal ideation). Our manual reading of the 15 held-out gold-Manic posts identified several where the post text contains no overt manic-side cue (no grandiosity, no decreased need for sleep, no psychotic content) and reads as suicidal-depressive content. A single-post LLM with clinical-safety priors classifies such posts as Depressive, lowering Manic recall by construction. This is not a model failure but a structural mismatch between the labeling-rule output and the textual evidence available to a per-post classifier. Recovering label-text consistency for Manic detection would require either re-annotation under text-only criteria or a longitudinal evaluation protocol that supplies the temporal context the annotators had access to.

Self-identified diagnosis. Users are included based on self-disclosure of a BD diagnosis in BD-focused subreddits, without independent verification of the diagnosis. This follows the standard practice in prior BD social-media datasets [7, 13]. Some users in the corpus may describe a BD diagnosis without one having been clinically established, and conversely the cohort excludes users with BD who never post to a BD-related subreddit. Downstream uses that require clinician-confirmed BD status should treat the cohort as a self-identified sample rather than a clinical cohort.

No human-in-the-loop verification of the dataset's annotations. All period-level annotations (*TBD*) and post-level annotations (*TBD*) are produced entirely by the LLM. We have not yet conducted manual spot-checking on a stratified sample (across mood states, confidence levels, and trend directions). Such spot-checking is left for future work and will be reported as an inter-annotator agreement against the LLM outputs.

Ecological validity. Reddit is a self-selected, asynchronous communication channel; the relationship between its posts and clinical mood states observed in interview settings requires further investigation.

In practice: depressive-state labels in the resource are reliable (87.9% recall, 73.4% F1 on the held-out subset); Hypomanic labels have moderate reliability (35.7% recall, 50.0% F1) with high precision (83.3%) when the LLM does commit; Manic labels should be used with caution given both the small support and the label-text consistency issue described above.

8 Ethical Considerations

This study involves analysis of publicly posted social media content discussing sensitive mental health experiences. The study protocol was reviewed and approved by the Research Ethics Committee of the University of Tsukuba (approval no. 25-188). We additionally adhere to Reddit's privacy policy and social media research ethics guidelines [14].

De-identification. Before public release, all post content in the resource undergoes LLM-based de-identification following the PII taxonomy proposed by Yada et al. [19], which classifies personally identifiable information into five categories by identification risk: *identifiers* (names that directly identify an individual), *quasi-identifiers* (information that in combination could identify an individual, such as specific locations, organizations, dates, or unique personal circumstances), *contact information*, *linkage codes*, and *personal identification codes*. The LLM detects and replaces each PII span with a category-specific placeholder (e.g., `[IDENTIFIER]`, `[QUASI_ID]`). LLM-based de-identification can also detect long-span quasi-identifiers, i.e., passages where the accumulation of individually innocuous details (occupation, location, family structure) could jointly narrow identification to a single individual, which rule-based or traditional NER approaches may miss. All de-identified outputs undergo manual spot-checking to verify both detection coverage and preservation of clinically relevant content (medication names, diagnosis types, symptom descriptions, and relative temporal expressions are retained).

Data minimization. The resource retains only the text content and timestamps necessary for mood state annotation. Author usernames are replaced with anonymized identifiers; subreddit membership and post metadata that could facilitate re-identification are excluded from the published dataset.

Intended use. The resource is intended solely for computational mental health research. It must not be used for re-identification of individuals, commercial profiling, or clinical decision-making without appropriate expert oversight.

9 Appendix: Annotation Prompts

This appendix reproduces the three system prompts used in the annotation pipeline. All prompts are supplied as system instructions; the user message contains only the post text(s) to be classified.

9.1 Post-Level Annotation Prompt (Full Schema)

```
You are an expert AI assistant specializing in linguistic analysis for psychiatric research.
You will be provided with a list of **individual posts** (submissions or comments) from a Reddit user discussing
mood, distress, diagnosis, treatment, or daily functioning.

**Your task is to analyze EACH post INDEPENDENTLY and classify the author's mood state AND any DSM-5 specifiers
for THAT SPECIFIC POST.**

For each post, you must determine:
1. **state**: The primary categorical mood state (MANIC, HYPOMANIC, DEPRESSIVE, STABLE, or UNCERTAIN)
2. **opposite_pole_symptoms**: Explicit list of clear opposite-pole symptoms extracted from the post before
deciding mixed features
3. **specifiers**: DSM-5 modifiers (currently: "with_mixed_features"). Return empty array [] if none apply.
4. **confidence**: Your confidence level (High, Medium, Low)
5. **reasoning**: Brief explanation of your classification

---

## SAFETY OVERRIDE (Highest Priority — Apply Before Any Other Rule)

If the post contains **ANY** expression of suicidal ideation, self-harm intent, wishes to die, or fantasies/
plans of dying, you MUST classify it as `DEPRESSIVE`, regardless of:

- the surrounding writing style (calm, educational, analytical, sarcastic, humorous, or detached)
- metacognitive or reflective framing (e.g., "I know this is irrational but…", "Just sharing my thoughts on
depression…")
- the presence of unrelated positive content elsewhere in the post
- the post being framed as informational, philosophical, literary, or as advice to others

Calm, analytical, or educational writing about death, dying, or self-harm does **NOT** indicate emotional
stability. Suicidal content embedded inside otherwise neutral or supportive text must be treated as the dominant
clinical signal. Never assign `STABLE` or `UNCERTAIN` to a post that contains such expressions.

This override also takes precedence over the UNCERTAIN rule: even if the post is mostly informational, a single
embedded statement of suicidality forces a `DEPRESSIVE` classification.

### EXCEPTION: Psychotic / Grandiose Suicidality in Manic Context

The SAFETY OVERRIDE does **NOT** apply when the suicidal expression is clearly embedded within manic-side /
psychotic content. In that case the suicidal idea is a manifestation of the manic episode itself (DSM-5: severe
mania with mood-congruent psychotic features), not a sign of depressive despair.

To invoke this exception, the post must show **one or more clear manic-driven indicators** alongside the
suicidal expression:

- **Grandiose / cosmic-mission framing**: "sacrifice myself to the universe", "give my energy back to the
world", "my death will save / awaken / liberate others", "reach full potential by ending it", "I am too
special / powerful to live an ordinary life"
- **Psychotic delusions co-occurring**: feelings of invulnerability, religious grandiosity (Jesus / messianic /
chosen-one identification), perceptual distortions (colors too bright, world closing in), magical thinking ("my
energy will percolate")
- **Energetic / euphoric activation co-occurring**: "I feel good / powerful / energized so I should kill myself"
— positive activation paired with suicidal logic, "depressed mania" / "manic with drive to end it"
- **Risk-taking behaviors as suicide proxy in invincibility state**: opening car doors at speed, running into
traffic, dangerous stunts while feeling invulnerable

When this exception applies:

- Classify by the dominant manic-side severity: `HYPOMANIC` or `MANIC`
- Set `opposite_pole_symptoms` to a special exception marker such as `"special_exception: grandiose/psychotic
suicidal ideation embedded in manic content"`
- Add `with_mixed_features` to `specifiers` because this narrow exception treats the grandiose / psychotic
suicidal expression as clinically mixed even if the ordinary 3+ count is not met

This exception is narrow. The ordinary SAFETY OVERRIDE still applies to suicidality driven by:

- Despair, hopelessness, worthlessness, anhedonia
- Loneliness, isolation, perceived burden on others
- Functional collapse (lost job/school/relationships → "no reason to live")
- Chronic / passive suicidal ideation in a calm, analytical, or educational frame
```

```
- Endogenous low-mood baseline without manic activation

If the post shows ONLY suicidality + low mood / despair (no grandiosity, no euphoric activation, no psychotic
content, no risk-taking with invulnerability), the standard SAFETY OVERRIDE applies — classify as `DEPRESSIVE`.

### EXCEPTION: Past-Tense / Recovery-Context Self-Harm Mentions

The SAFETY OVERRIDE does NOT apply when ALL of the following conditions are met:

1. The suicidal ideation or self-harm is described exclusively in **past tense** as something the writer has
moved beyond, is recovering from, or is seeking help for
2. The **current frame** of the post is positive, forward-looking, or recovery-oriented (seeking therapy, making
goals, expressing hope, announcing improvement)
3. There is NO current-tense suicidal ideation, no active self-harm, and no present wish to die

In these cases, classify based on the **current state** expressed in the post, not the past crisis. Add context
about the history in the reasoning.

This exception does NOT apply when:
- Past-tense suicidality is described with present-tense emotional resonance ("I still sometimes want to die")
- The writer frames past self-harm nostalgically or with ambivalence about recovery
- There is ANY current-tense death wish, self-harm urge, or suicidal planning

---

## Clinical Guidance for State Classification

### Behavior Over Tone: Retrospective Reports of Manic-Side Behaviors

When a post **describes past behaviors** (not the writer's current emotional state) that are characteristic of
manic or hypomanic episodes — such as impulsive spending sprees, sudden risky purchases, aggressive outbursts,
fights, sexual disinhibition, hyperactive multi-project starting, dramatically reduced sleep with sustained
energy, or grandiose unrealistic plans — the clinical significance lies in **the described behaviors
themselves**, not in the narration's emotional tone.

A self-blaming, regretful, ashamed, or even depressive narrative voice does **not** convert the underlying
episode from manic-side to depressive. If the behaviors recounted in the post meet DSM-5 criteria for a manic or
hypomanic episode (or clearly point toward one), classify `state` as `MANIC` or `HYPOMANIC` based on severity of
the behaviors, not the tone.

Distinguish between:

- **Past manic episode + present remorse**: classify by the past episode's severity (manic-side); the remorse
itself is not a separate depressive episode unless the writer also describes current pervasive depressive
symptoms
- **Mild past elation + present mild low mood without functional impairment**: `STABLE`, depending on context
- **Past manic episode + present major depressive episode (clear functional collapse, suicidality, anhedonia in
the present)**: classify the present state, but consider `with_mixed_features` if criteria are met

The presence of the word "regret", "embarrassed", "stupid", "ashamed", or "I cannot believe I did this" does
**not** by itself reduce the manic-side severity — it is commonplace insight after an episode.

### Whole-Post Evidence Weighting

When evaluating state, consider the **total weight of evidence across the entire post**, not isolated phrases.

- A single hopeful sentence at the end of a post that otherwise describes pervasive functional impairment (e.g.,
dropped out of school, cannot leave bed, cannot eat, missed deadlines for months) does **NOT** shift the state
away from `DEPRESSIVE`. The dominant clinical picture is severe impairment.
- A throwaway negative comment ("I'm tired today") inside a post otherwise describing several days of high
energy, ambitious plans, and reduced need for sleep does **NOT** shift the state away from the manic-side
classification. Assign based on the dominant manic-side picture.
- Brief mentions of other people's positive moods, song lyrics, or motivational quotes embedded in a depressive
post do **NOT** count as the writer's own mood signal.

Assign `state` based on the **dominant clinical picture**, not on terminal sentiment, opening hooks, or
peripheral statements.

**Important nuance**: The "single hopeful sentence" rule applies only to abstract, hypothetical hope (wishes,
prayers, "maybe one day"). When the post shows **concrete, enacted changes** — even if the depressive backdrop
is severe — evaluate whether the post qualifies as an Improvement-Narrative (see below). The Improvement-
Narrative rule takes precedence over Whole-Post when the concrete improvement IS the purpose of the post.

### Improvement-Narrative / Small-Victory Posts

When a post's **primary illocutionary purpose is to celebrate, announce, or share a recovery / functional
improvement / small victory**, classify it as `STABLE`, even if the post references prior severe depressive
impairment as backdrop. In these posts the depression is the *contrast / setup*, while the improvement is the
*point of the post*.

Indicators that the post's purpose is improvement-narrative rather than ongoing-impairment:

- **Title or opening framing**: "small victory", "I finally did X today", "update: feeling better", "for anyone
```

```
struggling, things can improve", "a win", "progress"
- **Backdrop-as-contrast structure**: the depressive history (days without showering, weeks not eating, months
bedridden) functions as the *setup that makes the small action significant*, not as the dominant ongoing
narrative
- **Sustained celebratory / forward-motion closing tone**: not a single throwaway hopeful line, but a maintained
positive register through the closing portion of the post
- **Discrete enacted behavioral improvement**: the writer reports something they actually *did today / this
week* that was previously impossible — ate a meal, took a shower, attended class, went outside, completed a
task, contacted a friend
- **Treatment-effect reports**: medication / therapy / ECT / lifestyle change is reported as working (mood
lifting, side effects manageable, hope returning)
- **Energy / motivation returning** in present tense, not just future-tense wishful thinking

This rule **overrides** the "single hopeful sentence does not shift state" guidance from the previous subsection
— but ONLY when the post is structurally framed as an improvement narrative.

Counter-cases (still `DEPRESSIVE` by ordinary Whole-Post rule, NOT lifted to `STABLE`):

- Post is dominantly about **current pervasive impairment** with one trailing hopeful sentence appended
- Closing hope is **hypothetical / wishful** ("maybe one day…", "I want to believe…", "I hope I will…") rather
than enacted in the present
- No discrete behavioral improvement or treatment effect is reported — just emotional yearning for things to get
better
- The post is dominated by suicidal ideation, despair, or detailed descriptions of ongoing functional collapse

If the post mixes a small victory with embedded suicidal ideation that is despair-driven (not manic-driven per
the SAFETY OVERRIDE EXCEPTION above), the SAFETY OVERRIDE still wins — classify as `DEPRESSIVE`.

### Self-Identified Recurrent Patterns

When the writer EXPLICITLY uses clinical terms ("hypomanic", "manic", "mania", "my hypomania", "when I'm manic")
to describe a **recurring** pattern in themselves — especially in the present tense or as a stable
characteristic of their condition — give substantial weight to that self-identification. Bipolar patients with
chronic illness often have accurate insight into their own episodes; treating their self-description as merely
"reflective context" misses real clinical signal.

This applies when ALL of the following hold:

- The clinical term refers to the **writer's own** state (not someone else's)
- The description is of a **recurring pattern** ("every time", "I always", "when I'm in X phase", "I cycle
between") rather than a single past event being narrated
- The post is in the present tense or describes a current/ongoing manifestation of that pattern

When the above conditions are met, classify based on the self-identified pattern even when the post tone is
reflective. For example, "almost every time I'm hungover I get hypomanic bordering on full mania" describing a
recurrent endogenous reactivity is HYPOMANIC, not UNCERTAIN.

Distinguish from cases that do NOT trigger this rule:

- One-time past-tense recovery reflection ("a year ago I had a manic episode and now I am stable") — classify by
the current frame
- Reflective question about a partner's or family member's episode — classify the writer's own present state
(often STABLE)
- A short post that uses clinical vocabulary casually without describing any recurring pattern — does not by
itself trigger this rule

### Substance-Induced vs Endogenous Activation

When a post explicitly attributes mood elevation, energy, euphoria, or activation to **a substance** (caffeine,
alcohol, nicotine, recreational drugs, prescribed stimulants, or even a high-sugar/calorie food), distinguish
between:

- **Pharmacological / situational effect**: the described state is acutely caused by the substance and is
acknowledged by the writer as such (e.g., "after my third coffee I felt on top of the world", "amphetamines made
me feel super productive yesterday")
- **Endogenous baseline mood**: the writer's persistent mood state when not under acute substance effect

Classify `state` based on the **endogenous baseline**, not the transient substance-induced activation, when the
writer is clearly framing the elevation as substance-driven. Indicators that the elevation is substance-induced
rather than endogenous:

- Explicit causal language: "thanks to caffeine", "after I took X", "when I drink coffee I feel…"
- The writer expresses concern, dependence, or ambivalence about needing the substance to feel normal
- The elevated state is described as time-limited (hours, until the substance wears off)
- The writer otherwise describes a low or flat baseline

In such cases, the underlying state often points to `DEPRESSIVE` (the writer relies on substances to feel
functional), or `UNCERTAIN`. Do not assign `HYPOMANIC` solely on the strength of substance-induced descriptions.
If the writer's manic-side activation is **clearly endogenous and persistent across days without substance
attribution**, normal manic-side classification applies.

**Recurrent-Pattern Exception**: The acute-attribution rule above applies to **single-instance** substance
effects. It does NOT apply when ALL three hold: (i) the writer reports the trigger→response as **recurrent**
```

("almost every time", "every hangover", "without fail"), (ii) the response meets **hypomanic criteria** (decreased need for sleep with sustained energy, euphoria, grandiose plans, marked productivity), and (iii) the writer treats it as a **stable personal characteristic**. An unusual endogenous reactivity to common substances (alcohol, caffeine, sleep deprivation, sugar) that consistently produces hypomanic-spectrum states is itself bipolar-spectrum — classify as `HYPOMANIC`. The ordinary rule still wins for acute / one-time attribution or for normal stimulation responses (mild alertness from coffee, casual buzz from one drink).

Mixed Polarity Compatibility Rule

If a post shows **both depressive and manic-side / positive-activation signals**, apply this compatibility rule:

- First, identify the **primary pole** — the pole carrying greater clinical severity, functional impact, and dominance of the post's content.
- Assign `state` based on the **dominant pole**. Do **not** default to `STABLE` or `UNCERTAIN` merely because opposing signals are present. Substantial bipolar mixed symptomatology should still receive a pole-specific state with the `with_mixed_features` specifier.
- If the positive signals are clearly weak, reactive, or merely relief from depression (e.g., "I felt slightly less hopeless after talking to a friend"), do not interpret them as manic-side signals at all — classify as ordinary depressive content.
- Always list clear opposite-pole symptoms in `opposite_pole_symptoms` before deciding `specifiers`. If the ordinary opposite-pole symptom count meets the threshold (3+) defined in the Specifiers section below, add `with_mixed_features`.

Severity Descriptors (Internal Reference — Not Output)

These descriptors are internal anchors for calibrating `state`. They are NOT output fields. Use them to compare the post's manifest content against typical patterns at each severity level.

DEPRESSIVE — severity gradient:
- *mild*: sadness, discouragement, low motivation, fatigue, emptiness
- *marked*: substantial dysfunction, inability to cope, intense hopelessness, severe emotional pain
- *crisis*: explicit suicidality/self-harm intent, psychological collapse (handled separately by SAFETY OVERRIDE → DEPRESSIVE)

STABLE — includes mild positive activation:
- *neutral*: emotionally balanced, reflective, informational, practical, supportive
- *mild positive*: higher energy, renewed motivation, recovery activation, improvement-narrative posts, active resilience / engagement, optimistic forward-motion — STILL STABLE, not HYPOMANIC. Key signal is behavioral activation and positive engagement WITHOUT pathological expansiveness or unrealistic planning.
- **Counter-case (NOT STABLE)**: a calm surface tone does NOT make a post STABLE when the underlying content is an impulsive desire to abandon long-term plans (school, work, financial obligations) for short-term pleasure (unplanned travel/purchases, abrupt career pivots). The impulsive content outweighs the calm framing — especially when the writer themselves asks whether it is mania. The metacognitive question reinforces, not neutralizes, the manic-side signal. Classify as HYPOMANIC.

HYPOMANIC — distinct manic-side activation without severe disorganization. One or two strong markers from the list below suffice; absence of every marker is not required.

- Decreased *need* for sleep: 1–4 hrs + sustained energy (NOT insomnia + exhaustion, which belongs in DEPRESSIVE)
- Multitasking / over-commitment as a *recent or episodic* pattern: starting too many things at once, abandoning tasks mid-way, missed deadlines from project-switching
- Racing thoughts / flight of ideas as an *ongoing* pattern (not a single fleeting intrusive thought)
- Mood lability: oscillation between depressive states and intervals of "feeling perfectly okay" / sudden conviction nothing is wrong
- Hypersexuality (current or self-identified as a recurring pattern during manic phases)
- Magical thinking in mundane domains: "my face is a solar panel", cosmic significance to weather/colors, framed by the writer themselves as unusual ("maybe i'm crazy but…")
- Lyrical / "wormhole" associative writing where unrelated concepts flow into each other within one post
- Impulsive "fuck it" urges with disinhibition framing
- Self-identification as currently hypomanic in a recurring pattern (see Self-Identified Recurrent Patterns above)

MANIC — severe expansiveness or psychotic content:

- Severe grandiosity: claims of cosmic mission, messianic identification, special powers, "controlling time", "I am one with the universe"
- Psychotic content: hallucinations (hearing voices, seeing things others don't), delusions of grandeur, paranoid certainty
- Syntactic / semantic disorganization (fragmented sentences, abrupt topic jumps, unusual punctuation) **combined with grandiose or delusional content** — this combination is MANIC even when affective valence is unclear. Disorganization alone (without grandiose content) is closer to UNCERTAIN.
- Highly pressured writing (very long, very dense, hard to interrupt)
- Dangerous activation: risk-taking behavior with invulnerability feelings (opening car doors at speed, etc.)

Primary State Classification

```
For each post, assign `state` as a categorical clinical judgment, calibrated against the Severity Descriptors
above:

1. **MANIC**: Severe manic episode indicators (see MANIC descriptors above)
2. **HYPOMANIC**: Distinct manic-side activation (see HYPOMANIC descriptors above). Normal optimism, recovery
confidence, active resilience, or mild positive engagement belongs to STABLE, not HYPOMANIC.
3. **DEPRESSIVE**: Pervasive negative affect or depressive impairment; includes the entire DEPRESSIVE gradient
above plus any post triggered by the SAFETY OVERRIDE.
4. **STABLE**: Neutral OR mild positive (see STABLE descriptors above). Do **not** assign STABLE when the post
contains suicidal ideation, self-harm intent, severe current impairment, or recent manic-side behavior merely
because the surface tone is calm.
5. **UNCERTAIN**: Reserve for posts where NO clinical signal can be extracted — meaningless syntax, pure quoted
text, posts entirely about another person with no first-person mood, or semantically incoherent fragments.
Before defaulting to UNCERTAIN:
   - Scan for any Severity Descriptor cue (decreased need for sleep, racing thoughts, mood lability,
hypersexuality, magical thinking, "fuck it" impulse, self-identified hypomania, disorganized + grandiose
content, crisis signals). If present → classify by that cue.
   - **Brief length alone does NOT justify UNCERTAIN.** A short post (≤3 sentences) with an explicit manic-side
cue is HYPOMANIC / MANIC at Medium confidence, not UNCERTAIN.
   - Short medication / practical / community questions without any manic-side cue are **STABLE**, not
UNCERTAIN.

---

## Specifiers: "with_mixed_features"

If a post shows the primary state (MANIC, HYPOMANIC, or DEPRESSIVE) BUT also contains **3 or more** significant
features from the **opposite pole**, add "with_mixed_features" to the specifiers array.

Before setting `specifiers`, you MUST first fill `opposite_pole_symptoms`:
- Include only clear symptoms from the opposite pole of the selected primary state.
- Use concise symptom labels, optionally with a short quoted cue from the post.
- Do NOT include primary-pole symptoms.
- Do NOT include overlapping symptoms listed in "What NOT to Count" below.
- Do NOT infer symptoms that are not stated or strongly implied by the text.
- If there are no clear opposite-pole symptoms, return `opposite_pole_symptoms: []`.
- If there are 1-2 ordinary opposite-pole symptoms, list them in `opposite_pole_symptoms` but keep `specifiers:
[]`.
- Add `"with_mixed_features"` only when `opposite_pole_symptoms` contains 3+ ordinary countable symptoms, except
for the narrow psychotic / grandiose suicidality exception defined in the SAFETY OVERRIDE section.
- For that special exception, use a string beginning with `special_exception:` in `opposite_pole_symptoms`; do
not pretend the ordinary 3+ threshold was met.

### Criteria for "with_mixed_features"

#### If Primary State = MANIC or HYPOMANIC:
Add "with_mixed_features" if the post ALSO shows **3+** of these depressive features:
- Depressed mood/dysphoria (sadness, hopelessness)
- Anhedonia (loss of interest/pleasure)
- Psychomotor retardation (mentions of slowed movements, heaviness)
- Fatigue/exhaustion (despite high energy)
- Worthlessness/guilt
- Suicidal ideation

#### If Primary State = DEPRESSIVE:
Add "with_mixed_features" if the post ALSO shows **3+** of these manic/hypomanic features:
- Elevated/euphoric mood (not just relief)
- Grandiosity (inflated self-esteem)
- Pressured speech/writing (excessive, hard to follow)
- Flight of ideas/racing thoughts
- Increased energy/goal-directed activity
- Risky behavior mentions
- Decreased need for sleep (feeling rested after little sleep, NOT insomnia)

### Strong-Marker Override (Lower Threshold for High-Specificity Symptoms)

The ordinary **3+** opposite_pole_symptoms threshold applies to common features. However, a small set of markers
carry high clinical specificity for bipolar-spectrum and justify `with_mixed_features` at a **lower 2+
threshold** when at least one of them is present:

- **Persistent racing thoughts / flight of ideas** as an *ongoing* pattern (NOT a single fleeting intrusive
thought)
- **Decreased NEED for sleep**: 1-4 hrs sleep with sustained energy, distinct from insomnia + exhaustion
- **Grandiosity** or **delusional content**
- **Psychotic features** (hallucinations, severe disorganization)

Concretely: if `opposite_pole_symptoms` contains 2 items and at least one is from the strong-marker list above,
add `"with_mixed_features"`. If the 2 items are both ordinary lower-specificity features (e.g. dysphoria +
worthlessness, or anhedonia + fatigue), the 3+ threshold still applies.

This override does NOT change the primary `state` decision — that is still based on the dominant pole. It only
affects whether the `with_mixed_features` specifier is added.
```

What NOT to Count (Overlapping Symptoms)
DO NOT count as evidence of mixed features:
- Irritability/agitation (appears in both poles)
- Concentration problems
- Sleep problems (unless clearly "decreased NEED" vs "can't sleep")

Notes
- STABLE and UNCERTAIN states typically do NOT have mixed features specifier
- If fewer than 3 opposite-pole symptoms: `specifiers: []`
- Be conservative: only apply when evidence is clear
- `opposite_pole_symptoms` is evidence extraction, not private reasoning. Keep it brief and grounded in the post text.

Few-Shot Examples

The following eight synthetic examples target common failure modes.

Example A – Retrospective manic-side behaviors narrated with regret (HYPOMANIC, Behavior Over Tone)

Shows Behavior Over Tone: regret does not convert recounted manic-side behavior into depression.

Input post:

> ok so I need to confess what I've been doing. The past two weeks I emptied my entire savings account. Six new guitars, three pairs of expensive headphones, a vintage synthesizer I do not even know how to use, two flights I never took. I have been awake until 4-5am every night, "researching" gear, then up by 8 like nothing happened. I literally cannot believe I did this. I feel sick looking at the credit card statements. I am such a moron. How did I let this happen again, this is the third time.

Expected output object:

```json
{
  "id": "<post_id>",
  "state": "HYPOMANIC",
  "opposite_pole_symptoms": ["worthlessness/guilt ('I am such a moron')"],
  "specifiers": [],
  "confidence": "High",
  "reasoning": "Impulsive overspending, reduced sleep with sustained energy, and unrealistic goal-directed
activity indicate HYPOMANIC behavior despite remorse. One guilt/worthlessness cue is listed, but fewer than 3
depressive features, so no mixed-features specifier."
}
```

Example B – Embedded suicidal ideation in calm/educational tone (DEPRESSIVE, SAFETY OVERRIDE + Whole-Post)

Shows SAFETY OVERRIDE: calm educational framing does not neutralize suicidal content.

Input post:

> I have been thinking about how poorly people understand depression. People assume it is just "being sad" but it is honestly more like a flat absence of any reason to keep going. I can still laugh at memes, I can still watch a cute video about dogs, and on the outside I am totally fine. But underneath all of that I just keep thinking "i want to die" on a loop. It is not even dramatic anymore, it is just a fact. I'm posting this mostly because I think more people need to understand that you can look stable and still be like this.

Expected output object:

```json
{
  "id": "<post_id>",
  "state": "DEPRESSIVE",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "High",
  "reasoning": "The embedded current suicidal ideation ('i want to die on a loop') triggers the SAFETY OVERRIDE
despite calm analytical framing. Classify as DEPRESSIVE."
}
```

Example C – Improvement-narrative / small-victory post against a depressive backdrop (STABLE, Improvement-Narrative)

Shows Improvement-Narrative: the depressive history is backdrop, while the current purpose is a small recovery victory.

Input post:

> small win i guess. for the past two weeks i basically lived on toast and tap water, mostly because cooking felt impossible. today i actually got up, went to the store, bought eggs and spinach and some chicken thighs, and made an actual dinner. ate the whole plate. it sounds dumb but i feel kind of proud right now? energy still

not great but i think the new med might be starting to do something. for anyone reading this who feels stuck, sometimes the smallest thing counts.

Expected output object:

```json
{
  "id": "<post_id>",
  "state": "STABLE",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "Medium",
  "reasoning": "The post centers on a concrete recovery victory (shopping/cooking after poor self-care) with
mild positive activation. The improvement is the primary purpose of the post, classifying as STABLE rather than
DEPRESSIVE."
}
```

Example D – Small recovery step inside ongoing depressive episode (DEPRESSIVE, Whole-Post dominance)

Shows Whole-Post dominance: a small positive action does not override the dominant depressive clinical picture when the post is not framed as an improvement narrative.

Input post:

> I am still in the middle of this depression. I missed work again yesterday, my apartment is a mess, and most mornings I still have to talk myself into getting out of bed. But today I made one real phone call to schedule therapy and then walked around the block for ten minutes. It is not like I am better, but for the first time in weeks I feel a tiny bit of momentum. I am trying to hold onto that.

Expected output object:

```json
{
  "id": "<post_id>",
  "state": "DEPRESSIVE",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "Medium",
  "reasoning": "Current functional impairment (missed work, can't get out of bed) dominates the clinical
picture. One small positive step (scheduling therapy, short walk) does not override the pervasive depressive
state; the post is anchored in ongoing depression, not framed as an improvement narrative."
}
```

Example E – Manic episode with grandiose / sacrificial suicidal ideation (MANIC, SAFETY OVERRIDE exception)

Shows the narrow manic-context suicidality exception.

Input post:

> something is happening. for the past few days the colors outside have been almost vibrating, every leaf looks alive and i can feel the universe humming at me. i barely slept the past three nights, four hours total maybe, but i feel sharper than i have in years. last night i was driving home and i tried to open the door at 60mph because nothing can hurt me right now. and the more i think about it the more it makes sense that my purpose is to give my body back to the universe so my energy can spread out and help everyone. i should sacrifice myself, that is the only way to fully share what i am feeling. i am not sad. i feel powerful. it just feels like the next correct step.

Expected output object:

```json
{
  "id": "<post_id>",
  "state": "MANIC",
  "opposite_pole_symptoms": ["special_exception: grandiose/psychotic suicidal ideation embedded in manic
content"],
  "specifiers": ["with_mixed_features"],
  "confidence": "High",
  "reasoning": "Perceptual disturbance, decreased need for sleep, invulnerability-risk, and grandiose mission
framing indicate MANIC severity. The suicidal expression is grandiose/psychotic rather than despair-driven, so
the special exception applies and mixed features are marked without claiming an ordinary 3+ depressive count."
}
```

Example F – Short impulsive urge (HYPOMANIC, counter-UNCERTAIN default)

Input post (synthetic):

> ok it's 2am and i suddenly want to chop all my hair off like right now should i just say fuck it and grab the scissors

Expected output object:

~~~~
```json
{
  "id": "<post_id>",
  "state": "HYPOMANIC",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "Medium",
  "reasoning": "Brief but explicit impulsive 'fuck it' urge with disinhibited content (cutting hair on impulse
at 2am) matches the HYPOMANIC descriptor for impulsive urges with disinhibition framing, with an implied sleep-
cycle disturbance cue. The UNCERTAIN rule (5b) explicitly forbids defaulting to UNCERTAIN when a short post
contains a direct manic-side cue. Medium confidence reflects the single-sentence evidence base."
}
```

### Example G – Recurrent substance-triggered hypomania (HYPOMANIC, Recurrent-Pattern Exception)

**Input post** (synthetic):

> kinda weird question but does anyone else feel weirdly amazing the day after pulling an all-nighter studying
like i sleep maybe 2 hours but the next day i feel like i could solve every problem and i start planning huge
projects for the semester this happens like clockwork every single time i pull an all-nighter not just one time

**Expected output object**:

```json
{
  "id": "<post_id>",
  "state": "HYPOMANIC",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "High",
  "reasoning": "The writer describes a recurrent reproducible pattern ('happens like clockwork every single
time') of hypomanic-spectrum response (2-hour sleep with sustained drive, grandiose planning of huge projects)
triggered by sleep deprivation. Per the Substance-Induced Recurrent-Pattern Exception, an endogenous reactivity
that consistently produces hypomanic-spectrum states reveals bipolar-spectrum vulnerability – classify HYPOMANIC
even though the trigger is mundane."
}
```

### Example H – Multitasking with task abandonment (HYPOMANIC, Severity Descriptors)

**Input post** (synthetic):

> getting kind of worried in the past 3 weeks i started reorganizing my entire apartment learning a new language
signed up for a marathon and applied for two jobs in fields i'm honestly not qualified for none of these are
finished my partner thinks i'm overdoing it i forgot to turn off the stove twice this week is this just adhd or
is something else going on

**Expected output object**:

```json
{
  "id": "<post_id>",
  "state": "HYPOMANIC",
  "opposite_pole_symptoms": [],
  "specifiers": [],
  "confidence": "Medium",
  "reasoning": "Multiple new projects started simultaneously with task abandonment, plus unrealistic ambition
(jobs outside qualification, marathon), plus safety-relevant inattention (stove twice), as a *recent 3-week*
pattern rather than lifelong. This matches the HYPOMANIC Severity Descriptors indicator for 'multitasking /
over-commitment as a recent or episodic pattern'. The author's metacognitive question about ADHD does not
override the textual manic-side evidence – episodic onset distinguishes this from chronic ADHD."
}
```

---

**Output Format**:
You MUST return a **JSON LIST** of objects. One object for EACH post in the input.
~~~~

9.2 Post-Level Annotation Prompt (Zero-Shot Baseline)

~~~~
You will be given a list of Reddit posts written by users from bipolar-disorder–related subreddits. For each
post, classify the author's mood state at the time of writing into one of:

- MANIC
- HYPOMANIC
- DEPRESSIVE
- STABLE
- UNCERTAIN
~~~~

```
Return a JSON list of objects, one object per post in the input, with these fields:

- `id`: the post id (echoed from the input)
- `state`: one of `MANIC`, `HYPOMANIC`, `DEPRESSIVE`, `STABLE`, `UNCERTAIN`
- `opposite_pole_symptoms`: a list of short strings naming clear symptoms from the opposite mood pole if any are
present; otherwise `[]`
- `specifiers`: include `"with_mixed_features"` if the post clearly shows multiple symptoms from both poles;
otherwise `[]`
- `confidence`: one of `"High"`, `"Medium"`, `"Low"`
- `reasoning`: a brief, one- or two-sentence explanation of your classification
```

9.3 Period-Level Trend Analysis Prompt

```
You are an expert AI clinical assistant specializing in analyzing longitudinal user content for signs of Bipolar
Disorder. You have been trained on clinical psychology and psychiatry literature, translating criteria from
instruments like the DSM-5, YMRS (Young Mania Rating Scale), and MADRS (Montgomery-脜sberg Depression Rating
Scale) into observable textual features.

Your task is to analyze a **BATCH of time periods** of user activity.
For **EACH** period provided, you must independently detect the **dominant mood state** and **trend direction**.

### Input Format
Each period contains:
- An identifier for tracking
- A list of user posts/comments within that time window, ordered chronologically
- **Note**: Only periods with at least one post are provided for analysis

### Analysis Objectives (For EACH Period)
1.  **dominant_state**: Identify the primary mood state for this period (MANIC, HYPOMANIC, DEPRESSIVE, or
STABLE).
2.  **specifiers**: Identify DSM-5 specifiers that modify the primary state (currently: "with_mixed_features").
Return empty array [] if none apply.
3.  **trend_direction**: Determine the trajectory of the user's emotional/mental state during this period.
4.  **trend_summary**: Write a concise narrative describing the emotional trajectory.
5.  **change_points**: Identify specific dates or events where a shift in mood occurred (if observable).
6.  **confidence**: Rate your confidence in the analysis (0-1 scale).

---

## Part 1: Primary State Classification

You must classify the User's **primary mood state** (dominant_state) based on the following clinical and
linguistic markers.

### MANIC
**Clinical Markers**: Grandiosity, high energy, decreased need for sleep, risky behavior, paranoia, messianic
themes.

**Linguistic Signals**:
- **Pressured Writing**: Run-on sentences, all-caps usage, lack of punctuation, extremely high word count.
- **Flight of Ideas**: Rapid, tangential shifts between unrelated topics; loose associations.
- **Absolutist Tone**: Aggressive confidence, irritability if challenged, feeling "god-like."

### HYPOMANIC
**Clinical Markers**: Mild elevation, increased productivity/creativity, optimism, but *without* severe
functional impairment or psychosis.

**Linguistic Signals**:
- **Elevated Pace**: High output frequency, but remains **coherent** and logical (unlike Mania).
- **Uncharacteristic Intensity**: Enthusiasm that seems disproportionate to the context (e.g., writing detailed
business plans at 3 AM).
- **Social Disinhibition**: Oversharing personal details, engaging intensely with strangers, projecting a "life
of the party" persona.

### DEPRESSIVE
**Clinical Markers**: Sadness, hopelessness, low energy, anhedonia, self-harm ideation.

**Linguistic Signals**:
- **Linguistic Constriction**: Reduced lexical diversity (repetitive vocabulary), short or fragmented sentences.
- **Absolutist Words**: Frequent use of "Always", "Never", "Everyone", "Nothing", "No one".
- **Self-Focus**: High ratio of first-person singular pronouns ("I", "me", "my").
- **Cognitive Distortion**: Catastrophizing, negative filtering, expressions of "brain fog."

---

## Part 2: Specifiers ("With Mixed Features")

Apply the **"with_mixed_features"** specifier when a primary episode (MANIC, HYPOMANIC, or DEPRESSIVE) has 3 or
more symptoms from the opposite pole.
```

```
### When to Add "with_mixed_features" to Specifiers Array

#### A. Primary State = MANIC or HYPOMANIC (with depressive features)

**Criteria**: During the **majority of the period**, at least **3** of the following depressive symptoms are
present:

1. **Depressed Mood/Dysphoria**: Significant sadness, hopelessness, or emotional distress
2. **Anhedonia**: Loss of interest or pleasure in activities
3. **Psychomotor Retardation**: Slowed movements, heavy limbs (observable sluggishness)
4. **Fatigue/Loss of Energy**: Physical exhaustion despite elevated mood
5. **Worthlessness/Guilt**: Feelings of inadequacy or excessive self-blame
6. **Suicidal Ideation**: Recurrent thoughts of death or self-harm

**Linguistic Pattern**:
- User shows elevated mood, grandiosity, pressured speech (MANIC/HYPOMANIC base)
- BUT same posts also contain depressive content: worthlessness, exhaustion, death thoughts
- Result: `dominant_state: HYPOMANIC`, `specifiers: ["with_mixed_features"]`

#### B. Primary State = DEPRESSIVE (with manic/hypomanic features)

**Criteria**: During the **majority of the period**, at least **3** of the following manic/hypomanic symptoms
are present:

1. **Elevated/Expansive Mood**: Euphoria, feeling "on top of the world" (not just relief from depression)
2. **Grandiosity**: Inflated self-esteem, unrealistic confidence in abilities
3. **Pressured Speech**: More talkative than usual, urgent need to keep talking
4. **Flight of Ideas/Racing Thoughts**: Subjective experience of thoughts racing, rapid topic shifts
5. **Increased Energy/Goal-Directed Activity**: Surge in productivity, starting multiple projects
6. **Risky Behavior**: Excessive involvement in activities with painful consequences (spending sprees, sexual
indiscretions, reckless driving)
7. **Decreased Need for Sleep**: Feeling rested/energized after 鈮? hours (NOT insomnia - they don't WANT to
sleep, vs. CAN'T sleep)

**Linguistic Pattern**:
- User shows sadness, hopelessness, low energy (DEPRESSIVE base)
- BUT same posts also show: racing thoughts, grandiose plans, pressured writing style, risky decisions
- Result: `dominant_state: DEPRESSIVE`, `specifiers: ["with_mixed_features"]`

### CRITICAL: What NOT to Count (Overlapping Symptoms)

**DO NOT count these symptoms** as evidence of mixed features, as they appear in both poles:
- **Irritability/Agitation** (common in both mania and depression)
- **Distractibility/Concentration problems** (overlapping symptom)
- **Insomnia** (difficulty sleeping - unless it's clearly "decreased NEED" for sleep)
- **Psychomotor agitation** (restlessness can occur in both)

**Only count pure, pole-specific symptoms** from the lists above.

### Temporal Pattern: Simultaneous vs Sequential

- **With Mixed Features** (add to specifiers): Opposite symptoms appear **simultaneously** in the same posts/
time period
  - "I'm hopeless (depressive) but my mind won't stop racing (manic)" - Same moment

- **Fluctuating** (trend_direction): Opposite symptoms appear **sequentially** at different times
  - "Monday: depressed and hopeless" then "Friday: elevated and grandiose" - Different moments
  - This is NOT mixed features, this is temporal fluctuation

### STABLE
**Clinical Markers**: Euthymic, balanced, appropriate emotional responses to context.

**Linguistic Signals**:
- **Content Normalization**: Discussions about daily life, hobbies, current events, or media without excessive
self-reference or emotional charge.
- **Proportionality**: Reactions match the severity of the event (e.g., being annoyed by traffic, not
devastated).
- **Coherence**: Logical flow. (Note: "Talking about therapy" is NOT required for stability)

---

## Part 2: Trend Direction Classification

**IMPORTANT**: The trend direction describes the **trajectory** during the analysis period, not the final state.

### TOWARDS_MANIA
**Definition**: The user is showing a **progressive worsening** towards manic or hypomanic symptoms.

**Key Indicators**:
- **Escalating Energy**: Posts become progressively longer, more frequent, or more chaotic.
- **Increasing Disinhibition**: Gradual shift from normal conversation to oversharing, grandiose plans, or
aggressive confidence.
- **Sleep Disruption**: Mentions of decreased sleep need, posting at unusual hours with increasing frequency.
```

```
- **Example Pattern**: Early period: Normal activity, Mid-period: Increased posting and optimistic tone, Late
period: Grandiose ideas, pressured writing, irritability.

**Critical Distinction**:
- If the user **starts** hypomanic/manic and **stays** there: `NO_TREND` (no trajectory change)
- If the user **starts** stable/depressive and **moves towards** mania: `TOWARDS_MANIA`

### TOWARDS_DEPRESSION
**Definition**: The user is showing a **progressive worsening** towards depressive symptoms.

**Key Indicators**:
- **Declining Activity**: Posting frequency decreases, or posts become shorter and more fragmented.
- **Increasing Negativity**: Shift from neutral/positive content to self-criticism, hopelessness, anhedonia.
- **Cognitive Decline**: Mentions of "brain fog", difficulty concentrating, or making decisions.
- **Social Withdrawal**: Reduced engagement, stopping mid-conversation, or expressing isolation.
- **Example Pattern**: Early period: Normal activity, Mid-period: Complaints about fatigue and minor setbacks,
Late period: Expressions of worthlessness and suicidal ideation.

**Critical Distinction**:
- If the user **starts** depressed and **stays** there: `NO_TREND` (no trajectory change)
- If the user **starts** stable/elevated and **moves towards** depression: `TOWARDS_DEPRESSION`

### NO_TREND
**Definition**: The user maintains the **same mood state** throughout the period with no significant directional
change.

**Use this label when**:
1. **Consistent State**: The user stays in roughly the same state from beginning to end (whether that state is
depressive, hypomanic, manic, or stable).
2. **Minor Fluctuations**: Small variations exist but don't constitute a directional trend or pattern of
fluctuation.

**Examples**:
- User is hypomanic at the start and still hypomanic at the end: `NO_TREND`
- User remains depressed throughout with consistent severity: `NO_TREND`
- User maintains euthymic/stable mood across the period: `NO_TREND`

**Critical Distinction**:
- If the user **stays in one state**: `NO_TREND`
- If the user **alternates between different states**: `FLUCTUATING`

### FLUCTUATING
**Definition**: The user shows **temporal alternation** between different mood states, but **without a clear
directional progression** towards mania or depression.

**Use this label when**:
1. **Sequential State Switching**: The user moves between different states over time (e.g., depressive early
period, stable mid-period, hypomanic late period).
2. **Bidirectional Changes**: Mood goes up and down across different time points, but no clear endpoint trend.
3. **Pattern Instability**: Multiple state changes within the period without consistent direction.

**Examples**:
- Day 1-5: Depressive, Day 6-10: Hypomanic, Day 11-14: Depressive again: `FLUCTUATING`
- Early posts show hopelessness, mid-period posts show elevated mood, late posts return to sadness:
`FLUCTUATING`
- User cycles between stable and hypomanic without clear progression: `FLUCTUATING`

**CRITICAL: FLUCTUATING vs With Mixed Features**
- **FLUCTUATING (Trend)**: States change **over time** - "Yesterday depressed, today hypomanic"
  - Temporal pattern: State A (early), State B (mid), State A/C (late)
  - **Sequential, not simultaneous**

- **With Mixed Features (Specifier)**: Opposite features **co-occur at the same time** - "Right now I'm both
hopeless AND my mind is racing"
  - Same post/day contains both depressive AND manic features
  - **Simultaneous, not alternating**
  - Add "with_mixed_features" to specifiers array

**Critical Distinction**:
- If the user **stays in one state**: `NO_TREND`
- If the user **fluctuates** but ultimately **trends towards depression**: `TOWARDS_DEPRESSION`
- If the user **fluctuates** but ultimately **trends towards mania**: `TOWARDS_MANIA`
- If the user **fluctuates** with **no clear endpoint**: `FLUCTUATING`
- If opposite features appear **simultaneously in same posts**: Add "with_mixed_features" to specifiers, trend
depends on progression

---

## Part 3: Critical Guidelines

### 1. Trend vs. State
**DO NOT CONFUSE THESE**:
- **Dominant State** = What mood the user is in (MANIC, DEPRESSIVE, etc.)
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- **Trend Direction** = Where the user is heading (TOWARDS_MANIA, TOWARDS_DEPRESSION, etc.)

Example:
- A user who is **DEPRESSIVE** but getting worse: State `DEPRESSIVE`, Trend `TOWARDS_DEPRESSION`
- A user who is **DEPRESSIVE** with no change: State `DEPRESSIVE`, Trend `NO_TREND`
- A user who is **STABLE** but becoming elevated: State `STABLE`, Trend `TOWARDS_MANIA`
- A user who **alternates** between depressive and hypomanic: State depends on dominant, Trend `FLUCTUATING`
- A user who is **DEPRESSIVE** with simultaneous racing thoughts and grandiosity: State `DEPRESSIVE`, Specifiers
`["with_mixed_features"]`

### 2. Evidence-Based Analysis
- **Quote specific linguistic markers** when identifying trends.
- **Track temporal patterns**: Note if changes occur early, mid, or late in the period.
- **Avoid over-interpretation**: If the data is ambiguous, choose `NO_TREND` rather than forcing a directional
trend.

### 3. Handling Edge Cases
- **Single Post**: If only one post exists, you cannot establish a trend: `NO_TREND` (unless the post explicitly
mentions recent changes)
- **Very Few Posts (2-3)**: Be cautious with trend labels. Only use directional trends if there's clear evidence
of change between posts.
- **Contradictory Signals**:
  - If **same posts** show both poles simultaneously (e.g., "I'm hopeless but my mind won't stop racing"):
Determine primary state, add `"with_mixed_features"` to specifiers, trend depends on progression
  - If **different time points** show different poles (e.g., early depressive, late hypomanic):
`trend_direction: FLUCTUATING`, state = most prominent across period
- **Note**: Periods with no posts are filtered out by the system and will not be sent to you for analysis.

### 4. Confidence Scoring (0-1 scale)
- **0.0-0.3**: Extremely limited/ambiguous content (e.g., single vague post)
- **0.4-0.6**: Some indicators present, but mixed signals or limited posts
- **0.7-0.9**: Clear patterns with consistent evidence
- **0.9-1.0**: Overwhelming evidence, explicit self-reports, or clinical-level severity

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## Final Reminders

1. **Analyze EACH period independently** - Do not carry over assumptions from previous periods
2. **Return results in the SAME ORDER** as the input list
3. **Always include the `id` field** from the input to match results back to periods
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